

Variational Inference and Bayesian Posterior Approximation

CAVI, SVGD, and Comparisons with MCMC

Ayush Mahale

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Abstract

One of the major challenges in statistics is approximating intractable Bayesian posteriors which are “difficult-to-compute probability densities.” Two methods that are foundational and very useful in tackling this problem are Variational inference and Markov Chain Monte Carlo (MCMC). MCMC is a sampling technique that constructs a stochastic process whose steady-state distribution is the posterior we are trying to sample from. After an initial burn-in time, we can treat the samples drawn from this chain as if they were drawn (approximately) from the posterior distribution. On the other hand, Variational Inference helps us approximate probability densities through optimization, seeking the member of a tractable family of densities that is closest to the true posterior in Kullback-Leibler (KL) divergence. When compared to MCMC, it turns out that variational inference is faster and more scalable, especially on massive data sets. In this essay, we will weave together insights from Blei *et al.* (2017) and Liu *et al.* (2016). We will explore Variational Inference in greater depth, including how it differs from MCMC, and two crucial Variational Inference algorithms: Coordinate Ascent Variational Inference (CAVI) and Stein Variational Gradient Descent (SVGD).

Typically, Bayesian models introduce latent variables z to capture the structure of the observed data x . In most cases when working with real data, the posterior $p(z|x)$ is intractable, which means that it cannot be written in a closed-form expression, forcing us to resort to approximation. One rather straightforward way to approximate the posterior is by MCMC. We begin MCMC by constructing an ergodic (proportion of time spent in each state is the probability of being in that state over time) Markov Chain, whose stationary distribution is the true posterior $p(z|x)$. By running the Markov Chain long enough, we can obtain samples from the Markov Chain that are approximately like sampling from the true posterior.

However, a major drawback of using MCMC arises when our posterior contains several well-separated modes. In this case, if we sample from a region of high density, our next sample is also likely to be from the same region. As a result, we rarely jump from one mode to another, and the chain gets stuck in the same region for a long time, missing other regions of the posterior which are also of high density. However in high dimensions, even if the posterior is unimodal, the MCMC can suffer from slow-mixing since most densities will be further from center and the chain will need to have a relatively small step-size, leading to inefficient exploration.

In direct contrast to MCMC, Variational Inference uses optimization instead of sampling. We start by positing a family $\mathcal{Q}$ of approximate densities $q(z)$. Then, we seek to find the member of the family $\mathcal{Q}$ that minimizes the KL divergence to the true posterior $p(z | x)$. Formally:

$$\begin{aligned} q^*(z) &= \arg \min_{q(z) \in \mathcal{Q}} \text{KL}(q(z) \parallel p(z | x)) \\ &= \arg \min_{q(z) \in \mathcal{Q}} \mathbb{E}_{q(z)} \left[\log q(z) - \log p(z | x) \right] \end{aligned} \tag{1}$$

It turns out that the KL divergence $\text{KL}(q | p)$ can be written as an Evidence Lower Bound (ELBO) plus a constant, and can therefore be optimized via gradient-based optimization through algorithms like CAVI and SVGD (*Blei, 2011, p.4*). Therefore, unlike in MCMC, we now have an optimization problem rather than sampling, making Variational Inference magnitudes faster and scalable to a large number of observations. However, it is important to realize that MCMC guarantees asymptotic exactness, which is not the case with Variational Inference where we get a close approximation of the posterior.

A key design choice for Inference typically is to minimize the reverse KL Divergence $\text{KL}(q | p)$ rather than forward KL Divergence $\text{KL}(p | q)$. $\text{KL}(q | p)$ is said to be “mode-seeking” (*Jones, 2021*), which means that the divergence tends to be low when q places density only where p places density, and high when q places density where p does not. Therefore, minimizing $\text{KL}(q | p)$ penalizes placing mass under q where p is negligible, but does not significantly penalize q missing modes of p . Hence, the reverse KL Divergence prefers q ’s that place significant mass in one mode of p rather than spreading q ’s mass across all modes of p . On the other hand, if we attempt to minimize the forward KL Divergence $\text{KL}(p | q)$, q would favor coverage of all modes of p , placing significant mass in regions of low densities that we typically do not care about. Additionally, since our true posterior p does not have a closed form expression, we cannot compute the expectation over p , but only approximate it via sampling which defeats Variational Inference’s goal of avoiding sampling. However, when computing the reverse KL Divergence, all the expectations $\mathbb{E}_q$ are taken with respect to q , as seen in (1). Unlike the true posterior p which is intractable, q is tractable by construction, and hence expectations under q can be computed.

One of the algorithms typically used for Variational Inference is the Coordinate Ascent

Variational Inference (CAVI) algorithm, applicable to conditionally conjugate models. First, we must assume a mean-field variational family where the “latent variables $[z_j]$ ’s are mutually independent” (*Blei et al., 2017, p.7*). A generic member of the mean-field variational family assumes the factorization:

$$q(\mathbf{z}) = \prod_{j=1}^m q_j(z_j) \quad (2)$$

Our goal with CAVI is to maximize the Evidence Lower Bound (ELBO):

$$\text{ELBO}(q) = \mathbb{E}_q[\log p(x, z) - \log q(z)] \quad (3)$$

Since $q(z)$ is factorizable by construction, we can write the ELBO as:

$$\begin{aligned} \text{ELBO}(q_1, \dots, q_m) &= \mathbb{E}_{q_1 \dots q_m}[\log p(x, z)] - \mathbb{E}_{q_1 \dots q_m}[\log(q_1(z_1) \cdots q_m(z_m))] \\ &= \mathbb{E}_{q_1 \dots q_m}[\log p(x, z)] - \sum_{i=1}^m \mathbb{E}_{q_i}[\log q_i(z_i)] \end{aligned} \quad (4)$$

When performing CAVI, we optimize ELBO with respect to one $q_j(z_j)$ at a time, while holding all other variational factors $q_\ell(z_\ell)$ (for $\ell \neq j$) fixed. Therefore, to update the j ’th factor, we must solve:

$$q_j^*(z_j) = \arg \max_{q_j} \text{ELBO}(q_1, \dots, q_j, \dots, q_m) \quad (5)$$

Setting the derivative to 0 allows us to find the optimal update for the j ’th factor up to proportionality:

$$q_j^*(z_j) \propto \exp\left\{\mathbb{E}_{q_{-j}}[\log p(z_j, z_{-j}, \mathbf{x})]\right\}, \quad (6)$$

where z_{-j} denotes every latent variable other than z_j .

It turns out that maximizing the ELBO over q is mathematically equivalent to minimizing $\text{KL}(q||p(z|x))$. In CAVI, the reason we choose to maximize the ELBO rather than minimize the KL Divergence is because the evidence $\log p(x)$ is difficult to compute. Let’s see why these two (maximizing ELBO and minimizing KL) are equivalent by starting with the definition of KL Divergence (*Blei et al., 2017, p.6*):

$$\begin{aligned} \text{KL}(q(z) || p(z | x)) &= \mathbb{E}_q[\log q(z)] - \mathbb{E}_q[\log p(z | x)] \\ &= \mathbb{E}_q[\log q(z)] - \mathbb{E}_q[\log p(z, x)] + \log p(x) \end{aligned} \quad (7)$$

From equation (3), we can see that the the KL Divergence is simply:

$$\text{KL}(q(z) || p(z | x)) = -\text{ELBO}(q) + \log p(x), \quad (8)$$

where $\log p(x)$ is a constant with respect to q and can therefore be ignored for optimization.

Therefore, each update is a descent step on the KL Divergence, since the ELBO (and hence the variational approximation) either increases or stays the same. We see that CAVI is nothing more than the regular Coordinate Ascent algorithm applied to the ELBO, where each variational factor is a “coordinate,” and we optimize the ELBO with respect to the current coordinate while holding all other coordinates fixed.

Notice that equation (6) is a valid update for any mean-field variational model, regardless of the form of the joint $p(z, x)$. It turns out that when each complete conditional (of the local variable z_i) $p(z_i | x, z_{-i}, \beta)$ belongs to the exponential family and the prior is conjugate, then a CAVI update produces a $q_i(z_i)$ that remains in the same exponential-family form as that complete conditional. This allows us to determine the optimal updates in closed form rather than up to proportionality (as seen in equation (6)).

We assume the form of the joint density $p(\beta, z, x)$:

$$p(\beta, z, x) = p(\beta) \prod_{i=1}^n p(z_i, x_i | \beta), \quad (9)$$

where “ β is a vector of global latent variables and z is a vector of local latent variables, whose j ’th component only govern data in the j ’th context” (*Blei et al., 2017, p.18*).

By construction, we know that z_i is conditionally independent of other local latent variables z_{-i} and other data x_{-i} . By construction again, the complete conditional has an exponential-family form. Therefore:

$$p(z_i | x_i, \beta, z_{-i}, x_{-i}) = p(z_i | x_i, \beta) = h(z_i) \exp\{\eta(\beta, x_i)^\top z_i - a(\eta(\beta, x_i))\}, \quad (10)$$

where “ z_i is its own sufficient statistic, $h(\cdot)$ is a base measure, $a(\cdot)$ is a log normalizer,” and η is the natural parameter of the complete conditional (*Blei et al., 2017, p.17*). It turns out that, by following the general CAVI update formula from equation (6) and plugging in $p(z_i, x_i | \beta)$ gives the local CAVI update:

$$\varphi_i = \mathbb{E}_{q(\beta|\lambda)}[\eta(\beta, x_i)] \quad (11)$$

As the prior on the vector of global latent variables β is conjugate to our likelihood kernel, it shares the same exponential-family form, with the natural parameter accounting for the data x . We can write the prior as:

$$p(\beta) = h(\beta) \exp\{\alpha^\top [\beta, -a(\beta)] - a(\alpha)\}, \quad (12)$$

Multiplying by the (point-wise) likelihood $p(z_i, x_i | \beta)$ gives us the complete conditional of β :

$$p(\beta | z, x) \propto h(\beta) \exp\left\{[\alpha_1 + \sum_{i=1}^n t(z_i, x_i), \alpha_2 + n]^\top [\beta, -a(\beta)]\right\} \quad (13)$$

Now, we can simply apply the generic coordinate update (6) and take the expectation over the “variational posteriors” $q(z_i | \varphi_i)$. This replaces the sufficient statistics $t(z_i, x_i)$ by their expectations, giving us the global variational update:

$$\lambda = [\alpha_1 + \sum_{i=1}^n \mathbb{E}_{q(z_i | \varphi_i)}[t(z_i, x_i)], \alpha_2 + n]^\top \quad (14)$$

leading to $q(\beta | \lambda)$ staying in the same exponential-family form as (13).

Now, the local and global updates are:

$$\varphi_i = \mathbb{E}_{q(\beta | \lambda)}[\eta(\beta, x_i)], \quad \lambda = [\alpha_1 + \sum_{i=1}^n \mathbb{E}_{q(z_i | \varphi_i)}[t(z_i, x_i)], \alpha_2 + n]^\top,$$

which alternately (update each φ_i with the current λ , and then update λ with all the current $\varphi_1, \dots, \varphi_n$) increase the ELBO until we reach convergence. Since each step is non-decreasing with regards to the ELBO, it can equivalently be seen as a descent step on the KL Divergence.

While the mean-field assumption is very convenient for updating the factors, we must assume that there are no posterior dependencies among the latent variables z_j ’s. Since this is not usually true, we systematically fail to capture any correlations between the latent variables.

Consider the case where we have two latent variables $z = (z_1, z_2)$ and a posterior $p(z | x)$ that we cannot directly sample from. We start by assuming a mean-field family factorization $q(z) = q_1(z_1) q(z_2)$, and our aim is to minimize the reverse KL Divergence: $\text{KL}(q(z) || (p(z | x)))$. In our example, the true posterior is unimodal in two highly correlated variables. Formally,

$$p(z | x) = \mathcal{N}(0, \Sigma), \quad \Sigma = \begin{pmatrix} \sigma^2 & \rho \sigma^2 \\ \rho \sigma^2 & \sigma^2 \end{pmatrix}, \quad |\rho| = 0.9 \quad (15)$$

Since z_1 and z_2 are highly correlated, if z_1 is large (either positive or negative), z_2 is also likely to be of similar magnitude (and sign). Therefore, a relatively amount of mass lies in directions perpendicular to the diagonal $z_1 = z_2$. This means that the true posterior density lies in a narrow elliptical shape with very thin tails. However, by construction of our mean-field family, our covariance matrix is diagonal, and our density approximations are therefore circles (unlike the elliptical shape of the true posterior).

Our goal with CAVI is to minimize the KL Divergence. Therefore, CAVI will pick the circle that approximates the true posterior (ellipse) best (i.e. has the lowest KL Divergence). It turns out that, when considering $\text{KL}(q || p)$, if q places density where p has relatively low density, then the KL Divergence increases significantly (heavy penalty). However, if q fails to cover regions where p has high density, the penalty on the KL Divergence is a lot smaller. Therefore, CAVI selects the smallest circle that roughly sits inside the ellipse (the true posterior), and will almost never engulf the posterior since the thin tails of the posterior

will add a heavy penalty on the KL Divergence. Therefore, in examples like these where we are working with strongly correlated variables, CAVI consistently underestimates the marginal variances. This is because CAVI is biased to prefer underestimating rather than overestimating density, and it cannot capture the marginal variance in each variable while assuming independence without overestimating density.

However, despite these biases, the variational distribution produced by mean-field is still very useful. For example, it can be used to address a major limitation of Markov Chain Monte Carlo (MCMC). While MCMC methods let us draw samples as if they were sampled from posterior (which is useful when working with intractable distributions) we can often get stuck at local optima. Our goal, however, is to sample from regions of high density to begin with, which is the global optimum. However, the burn-in period to converge to a global optimum can be quite long due to getting stuck at these local optima.

By construction in mean-field, q places most of its mass near a single true posterior mode (since it minimizes KL Divergence). Therefore, we can treat the q from mean-field as an initial proposal (i.e. draw initial samples from it) when performing MCMC, which prevents us from getting stuck at local optima, dramatically reducing the burn-in period.

Another Variational Inference algorithm is Stein Variational Gradient Descent (SVGD). SVGD is a particle-based algorithm that approximates the target posterior p by iteratively updating a set of particles to minimize the KL Divergence in a reproducing kernel Hilbert space (RKHS).

We will start by defining the Stein operator applied to a vector-valued function $\phi : \mathbb{R}^d \rightarrow \mathbb{R}^d$ as follows:

$$\mathcal{T}_p^\phi f(x) = \phi(x)^\top \nabla_x \log p(x) + \nabla_x \cdot \phi(x) \quad (16)$$

Then, select a positive-definite kernel ($k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ and $k(x, x') = k(x', x)$) with a reproducing kernel Hilbert Space $\mathcal{H}$ with feature map ψ satisfying the following:

$$k(x, x') = \langle \psi(x), \psi(x') \rangle_{\mathcal{H}} \quad (17)$$

Note that Liu and Wang in their study present the RBF kernel as a good candidate for SVGD.

Now, our goal with each update is to minimize the KL Divergence between q and p , and it turns out that the steepest descent direction ϕ^* restricted to the unit ball in $\mathcal{H}^d$ is:

$$\phi^*(\cdot) = \mathbb{E}_{x \sim q} [k(x, \cdot) \nabla_x \log p(x) + \nabla_x k(x, \cdot)] \quad (18)$$

where:

1. Attraction term: $k(x, \cdot) \nabla_x \log p(x)$. This term attracts the particles to regions of high density (in the true posterior).

2. Repulsion term: $\nabla_x k(x, \cdot)$. This term pushes particles away from each other so that particles “spread out” and don’t accumulate at a single mode of the true posterior. Therefore, unlike in MCMC where the random proposals lead to variance in samples, in SVGD, the kernel gradient is entirely responsible for the variance in samples.
3. ϕ^* . This term is the “functional gradient” of the KL in the RKHS.

We then iteratively update the particles as follows:

$$x_i^{\ell+1} \leftarrow x_i^\ell + \epsilon_\ell \hat{\phi}^*(x_i^\ell), \text{ where } \hat{\phi}^*(x) = \frac{1}{n} \sum_{j=1}^n \left[k(x_j^\ell, x) \nabla_{x_j^\ell} \log p(x_j^\ell) + \nabla_{x_j^\ell} k(x_j^\ell, x) \right], \quad (19)$$

where ϵ_ℓ represents the step-size at the ℓ ’th iteration. Note that we approximate the expectation under q as seen in equation (18) (when computing ϕ^*) with the empirical mean of the particles at the ℓ ’th iteration.

SVGD, therefore, optimizes family of distributions that can be represented empirically:

$$\mathcal{Q} = \left\{ q(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{x = x_i\} \mid n \in \mathbb{N}, x_i \in \mathbb{R}^d \right\} \quad (20)$$

We notice that each x_i is a free parameter, and so the family is non-parametric. This means that increasing n lets q approximate any continuous posterior arbitrarily closely (*Liu, Wang, 2016, p.5*). This is the key difference from mean-field, whose independence assumption and therefore product form has limited flexibility and cannot represent correlations in variables or multiple modes.

Another ensemble method we looked at in class is Parallel Tempering. In Parallel Tempering, we run multiple MCMC chains in parallel, each at a different temperature β_k (*Earl, Deem, 2024, p.2*). Chain k targets (stationary distribution of the chain) the tempered distribution:

$$p_k(x) \propto p(x)^{\beta_k}, \quad (21)$$

where $\beta_k \in (0, 1]$.

Each chain performs Metropolis-Hastings updates targeting that chain’s tempered density $p(x)^{\beta_k}$. Parallel Tempering achieves good sampling since chains at different temperatures are allowed to “exchange complete configurations” (*Earl, Deem, 2024, p.2*). High temperature states typically sample large volumes of phase space, then swap those “exploratory” states down to the low temperature states (which have precise sampling in local regions).

Let’s now explore bandwidth selection. In their experiments, Liu and Wang use the RBF Kernel:

$$k(x, x') = \exp\left(-\frac{1}{h} \|x - x'\|_2^2\right), \quad (22)$$

setting the kernel bandwidth via the “median trick” as:

$$h = \frac{\text{med}^2}{\log n}, \quad (23)$$

where med is the “median of the pairwise distance between the current points $\{x_i\}_{i=1}^n$.” This heuristic aims to balance out attraction and repulsion for each x_i :

$$\sum_j k(x_i, x_j) \approx n \exp\left(-\frac{1}{h} \text{med}^2\right) = 1, \quad (24)$$

automatically adapting h across the iterations. This “median trick” works well in low dimensions because we get a bandwidth that is roughly within a constant factor of the typical distance between two particles drawn from the current posterior. In this case, SVGD’s attraction and repulsion will remain roughly equal for each x_i . However, in higher dimensions, the distance between two random x_i ’s concentrate around their mean. Then, the sum of the kernel weights will scale constantly, and the repulsion term goes to zero. Therefore, it is no longer the case that attraction and repulsion stay roughly equal, but instead the attraction term scales with the mean, and all the particles continuously move in the same direction.

The “median-trick” is usually applied once before we start SVGD. As particles move, the h we chose can turn out to be a bad choice. However, we can optimize this procedure by adapting the value of h after every iteration of the SVGD algorithm. As an example, after each iteration, update h to be one of the candidates in the set $\{h, \frac{h}{2}, 2h\}$ based on which choice of h gives the smallest KL Divergence.

Another core idea in SVGD is the “functional gradient” of the KL Divergence. The functional gradient of $\text{KL}(q|p)$, tells us, for each point, how we should perturb our density q to decrease the KL Divergence as fast as possible. As per Theorem 3.1 in Liu & Wang, functional gradient can be written as:

$$\nabla_\epsilon \text{KL}(q_{[T]} \| p) \big|_{\epsilon=0} = -\mathbb{E}_{x \sim q} [\text{trace}(\mathcal{A}_p \phi(x))], \quad (25)$$

where $\mathcal{A}_p \phi(x) = \nabla_x \log p(x) \phi(x)^\top + \nabla_x \phi(x)$ is the Stein operator.

We can interpret ϕ as the direction in which we perturb the density q . The functional gradient of $\text{KL}(q|p)$ at q is the velocity field ϕ^* that maximizes:

$$\langle \nabla_\phi \text{KL}, \phi \rangle = -\mathbb{E}_q [\text{trace}(\mathcal{A}_p \phi)] \quad (26)$$

Maximizing ϕ to lie in the “unit ball of a RKHS” (*Liu, Wang, 2016, p.2*) gives us the closed form solution for ϕ^* :

$$\phi^*(\cdot) = \mathbb{E}_{x \sim q} \left[k(x, \cdot) \nabla_x \log p(x) + \nabla_x k(x, \cdot) \right], \quad (27)$$

which is the functional steepest descent on the KL Divergence. We see that this quantity (functional gradient of KL Divergence) is calculated using only computationally available quantities, since:

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1. $\nabla_x \log p(x)$ is simply the derivative of the log of our (tractable) target density. Since we are working with the log of the target density, the target density can be unnormalized.
 2. $k(x, \cdot)$ and $\nabla_x k(x, \cdot)$ only depend on our choice of the kernel (RBF in our case).

Ultimately, our goal is to approximate probability densities which are difficult to compute. The choice between MCMC, CAVI, and SVGD depends on the nature of the problem. If we are working with limited data but seek asymptotic exactness instead of rough approximation, then we would choose MCMC due to sampling being convenient with limited data. However, if we have large amounts of data, sampling is not always feasible, and we must resort to inference and optimization through methods like CAVI (mean field Variational Inference) and SVGD (particle Variational Inference). The difference between these two approaches (sampling vs inference) boils down to a trade-off between accuracy (offered by MCMC) and speed (offered by Variational Inference).

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